stream along with one another they can produce water waves, with the attendant interference properties illustrated in Figure 4.7. And so, it might seem reasonable to guess that wave properties, such as interference patterns, can arise from a particle picture of light provided a huge number of photons, the particles of light, are involved.

In reality, though, the microscopic world is far more subtle. Even if the intensity of the light source in Figure 4.8 is turned down and down finally to the point where individual photons are being fired one by one at the barrier—say at the rate of one every ten seconds—the resulting photographic plate will still look like that in Figure 4.8: So long as we wait long enough for a huge number of these separate bundles of light to make it through the slits and to each be recorded by a single dot where they hit the photographic plate, these dots will build up to form the image of an interference pattern, the image in Figure 4.8. This is astounding. How can individual photon particles that sequentially pass through the screen and separately hit the photographic plate conspire to produce the bright and dark bands of interfering waves? Conventional reasoning tells us that each and every photon passes through either the left slit or the right slit and we would therefore expect to find the pattern shown in Figure 4.6. But we don’t.

If you are not bowled over by this fact of nature, it means that either you have seen it before and have become blasé or the description so far has not been sufficiently vivid. So, in case it’s the latter, let’s describe it again, but in a slightly different way. You close off the left slit and fire the photons one by one at the barrier. Some get through, some don’t. The ones that do create an image on the photographic plate, dot by single dot, which looks like that in Figure 4.4. You then run the experiment again with a new photographic plate, but this time you open both slits. Naturally enough, you think that this will only increase the number of photons that pass through the slits in the barrier and hit the photographic plate, thereby exposing the film to more total light than in your first run of the experiment. But when you later examine the image produced, you find that not only are there places on the photographic plate that were dark in the first experiment and are now bright, as expected, there are also places on the photographic plate that were bright in your first experiment but are now dark, as in Figure 4.8. By increasing the number of individual